PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

Date: 19 June 2026

Team ID: -----

Project Name: AI-Enhanced Intrusion Detection System

Maximum Marks: 4 Marks

Data Flow Diagram (DFD)

DFD Level 0

```
+------------------+
|      User        |
+------------------+
         |
         | Input Network Features
         v
+-----------------------------+
| AI-Enhanced IDS Application  |
+-----------------------------+
         |
         | Process Features
         v
+-----------------------------+
| Data Preprocessing Layer     |
| (StandardScaler)             |
+-----------------------------+
         |
         | Scaled Data
         v
+-----------------------------+
| Random Forest ML Model       |
+-----------------------------+
         |
         | Prediction Result
         v
+-----------------------------+
| Attack Classification        |
| Normal / DoS / Probe /       |
| R2L / U2R                    |
+-----------------------------+
         |
```

```
         +----------------+
         |                |
         v                v
+----------------+  +------------------+
| Alert Log      |  | Dashboard & API  |
+----------------+  +------------------+
```

DFD Level 1

```
User
 |
 | Enter Network Traffic Features
 v
[Prediction Module]
 |
 | Feature Validation
 v
[Data Preprocessing]
 |
 | StandardScaler
 v
[Machine Learning Engine]
 |
 | Random Forest Classification
 v
[Prediction Result]
 |
 +-------------------------+
 |                         |
 v                         v
[Alert Management]    [Dashboard]
 |
 v
[Prediction Logs]
```

Data Flow Description

1. User enters network traffic feature values through the web interface.
2. Input data is validated by the Flask application.
3. Features are normalized using StandardScaler.
4. The Random Forest model processes the input.
5. The model predicts one of five traffic classes.
6. Results are displayed on the dashboard.
7. Attack alerts are logged for monitoring.
8. REST API provides programmatic access to prediction results.

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Security Analyst	Intrusion Detection	USN-1	As a security analyst, I can submit network traffic data for analysis.	System accepts feature inputs and processes them successfully.	High	Sprint-1
Security Analyst	Attack Classification	USN-2	As a security analyst, I can view attack classifications generated by the model.	Predicted attack category is displayed correctly.	High	Sprint-1
Security Analyst	Confidence Score	USN-3	As a security analyst, I can view prediction confidence scores.	Confidence percentage is shown with prediction results.	High	Sprint-1
Security Analyst	Dashboard Monitoring	USN-4	As a security analyst, I can monitor intrusion statistics on a dashboard.	Dashboard displays attack statistics and charts.	High	Sprint-2
Security Analyst	Alert Management	USN-5	As a security analyst, I can view all detected attack logs.	Alert records are stored and displayed correctly.	High	Sprint-2
Network Administrator	API Access	USN-6	As a network administrator, I can access predictions through REST APIs.	API returns valid prediction responses.	Medium	Sprint-3
Network Administrator	Threat Monitoring	USN-7	As a network administrator, I can monitor suspicious	System updates alerts and predictions	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
			network behavior in real time.	continuously.		
Administrator	System Management	USN-8	As an administrator, I can manage system deployment and maintenance.	Application services run successfully without interruption.	Medium	Sprint-4
Administrator	Model Updates	USN-9	As an administrator, I can retrain and update the machine learning model.	Updated model is deployed successfully.	Medium	Sprint-4
Administrator	Security Reporting	USN-10	As an administrator, I can generate reports of detected threats.	Reports are generated with attack summaries and statistics.	High	Sprint-4

Conclusion

The Data Flow Diagram illustrates how network traffic data moves through the AI-Enhanced Intrusion Detection System, from user input to attack classification and alert generation. The user stories define the functional requirements of security analysts, network administrators, and system administrators, ensuring that the application meets cybersecurity monitoring and intrusion detection objectives.